

KARACHI: A SPATIAL— TEMPORAL CRIME INTELLIGENCE SYSTEM USING GEOSPATIAL CLUSTERING AND PROBABILISTIC RISK PROFILING

Machine Learning
Project Report

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A Spatial–Temporal Crime Intelligence System using Geospatial Clustering and Probabilistic Risk Profiling

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Summary

Urban crime analysis requires models that capture complex spatial and temporal patterns while remaining interpretable and ethically sound. This project presents a Spatial–Temporal Crime Intelligence System that applies unsupervised machine learning, geospatial clustering, and probabilistic risk profiling to analyze historical crime data from Karachi and infer location-based crime risk.

The primary objective of the system is not to predict individual criminal events, but to provide risk-aware spatial intelligence by segmenting the city into meaningful geographic regions and analyzing historical crime behavior within those regions. This approach ensures interpretability, robustness, and practical relevance.

The dataset was first cleaned and standardized. Semantically similar crime categories were normalized into unified classes, non-informative identifiers were removed, and spatial attributes (latitude and longitude) were retained. Temporal attributes were processed to support monthly aggregation for trend analysis.

Spatial segmentation was performed using both centroid-based and density-based clustering techniques. K-Means clustering was evaluated using the Silhouette Coefficient to determine the optimal number of clusters, achieving a maximum silhouette score of approximately 0.58, indicating well-separated spatial regions. In parallel, density-based clustering (DBSCAN / OPTICS) was applied to identify spatial crime hotspots without predefining the number of clusters. A fallback strategy ensured robustness across execution environments.

Each spatial cluster was then profiled using historical data to infer the dominant Town and Subdivision, along with probability distributions over the top geographic areas. In addition, empirical probability distributions of crime types were computed for each cluster, allowing the system to report the most and least likely crimes for any given location based on historical frequency.

To capture temporal dynamics, monthly crime counts were aggregated at the cluster level and modeled using linear regression, enabling short-term projection of total crime activity trends. This provided interpretable insight into whether crime levels within a region were increasing or decreasing over time.

For real-time inference, unseen geographic coordinates are assigned to the nearest labeled spatial cluster using a nearest-neighbor mapping strategy. The system then returns the likely Town and Subdivision, historical crime-type likelihoods, and temporal trend projections where sufficient data exists.

1. Introduction

Urban crime analysis poses significant challenges due to its strong dependence on geographic location, temporal variation, and administrative boundaries. With the increasing availability of spatial data, machine learning techniques offer an opportunity to move beyond descriptive analysis and toward intelligent systems capable of providing interpretable, location-based insights. However, traditional supervised crime prediction models often struggle with data imbalance, overfitting, and ethical concerns when applied to public safety contexts.

The purpose of this project is to design and implement a Spatial–Temporal Crime Intelligence System that analyzes historical crime data using unsupervised machine learning techniques. The project aims to infer area-level crime risk and patterns based on geographic coordinates rather than predicting individual criminal incidents. The system is intended to support exploratory analysis, spatial risk awareness, and decision support through interpretable and responsible modeling.

1.1. Scope of the Project

The scope of this project focuses on the development of an end-to-end machine learning system for spatial–temporal crime intelligence. Specifically, the project includes:

- ✓ Preprocessing and normalization of historical crime data, including consolidation of semantically similar crime categories
- ✓ Spatial segmentation of the city using latitude and longitude through unsupervised clustering
- ✓ Profiling of spatial clusters to infer dominant Towns and Subdivisions
- ✓ Estimation of historical crime-type likelihoods within each spatial cluster
- ✓ Temporal analysis and projection of total crime volume at the cluster level
- ✓ Real-time inference for unseen geographic coordinates using nearest-neighbor assignment
- ✓ Development of an interactive application for visualization and user interaction

The project does not attempt to predict individual crimes, identify offenders, or establish causal relationships. All outputs are framed as historical risk indicators, ensuring ethical and responsible use.

1.2. Development of the Project

The project was developed through a structured, iterative process. Initial stages focused on data cleaning and feature standardization to ensure consistency across spatial and categorical attributes. Unsupervised clustering techniques were then applied to identify natural geographic regions within the city. Multiple clustering approaches were evaluated, including centroid-based and density-based methods, using appropriate internal validation metrics such as the Silhouette Coefficient and cluster density characteristics.

Following spatial segmentation, each cluster was analyzed to extract meaningful geographic and crime-related profiles. Probabilistic crime-type distributions were computed to characterize historical risk patterns, while temporal aggregation enabled trend projection using regression modeling. Finally, all analytical components were integrated into a deployment-ready inference pipeline, with trained models and artifacts serialized for reuse.

An interactive web-based application was implemented to demonstrate the system's practical usability, allowing users to query locations and visualize spatial risk information in real time.

1.3. Thesis of the Project

The central thesis of this project is that unsupervised spatial-temporal modeling provides an effective and interpretable foundation for urban crime intelligence systems. By combining geospatial clustering, probabilistic risk profiling, and temporal trend analysis, the project demonstrates that meaningful location-based crime insights can be derived without relying on supervised prediction models. The resulting system highlights the value of responsible, data-driven approaches for spatial intelligence in complex urban environments.

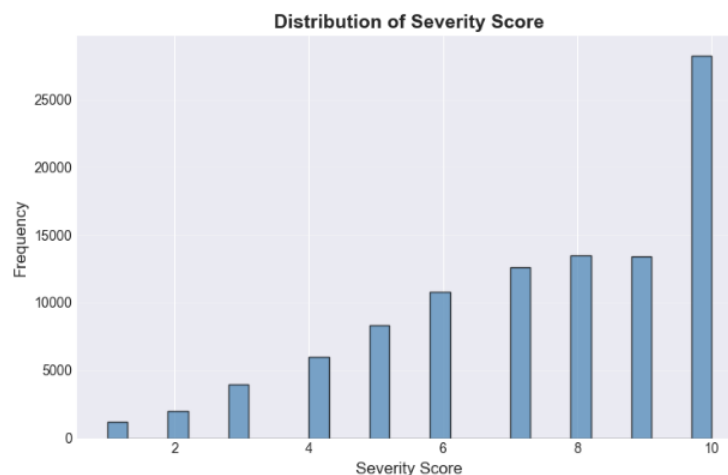
2. Exploratory Data Analysis (EDA)

The objective of this analysis is to identify dominant patterns, assess class imbalance, and gain intuition about the underlying structure of the data that may influence clustering and risk profiling.

2.1 Distribution of Severity Score

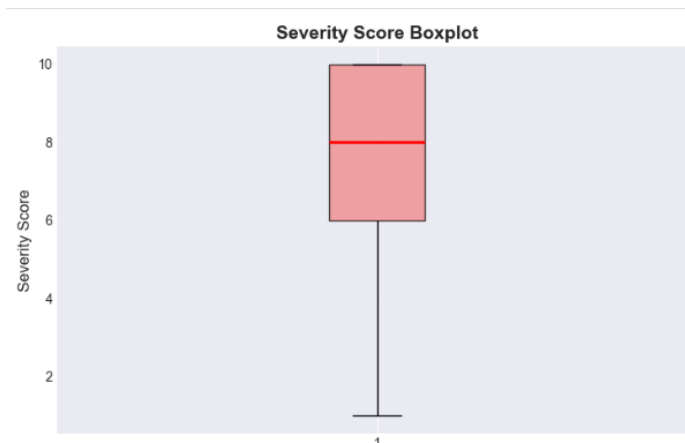
The distribution of the Severity Score shows a strong concentration toward higher values. As illustrated in the histogram, the frequency of incidents increases steadily from lower severity scores to higher ones, with the largest number of incidents occurring at the maximum severity level.

This indicates that a substantial portion of recorded incidents are classified as high-severity events, suggesting that the dataset is skewed toward more serious crimes rather than minor offenses.



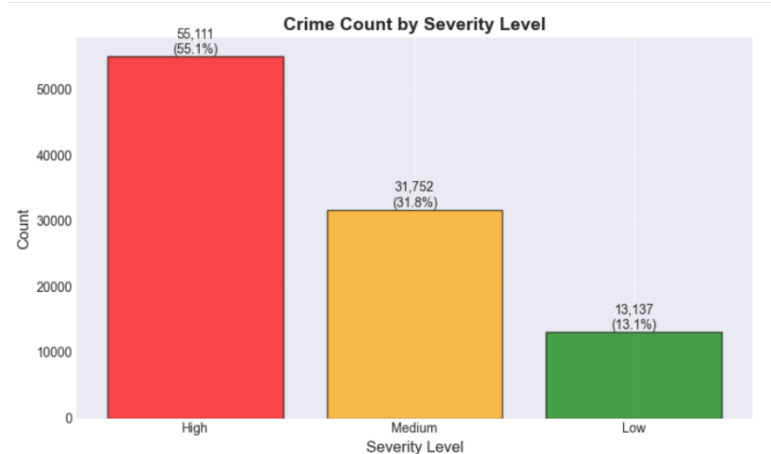
2.2 Severity Score Boxplot

The boxplot of severity scores further confirms the skewed nature of the data. The median severity score is relatively high, with the interquartile range concentrated in the upper portion of the scale. Lower severity values appear as outliers, indicating that less severe crimes occur less frequently.



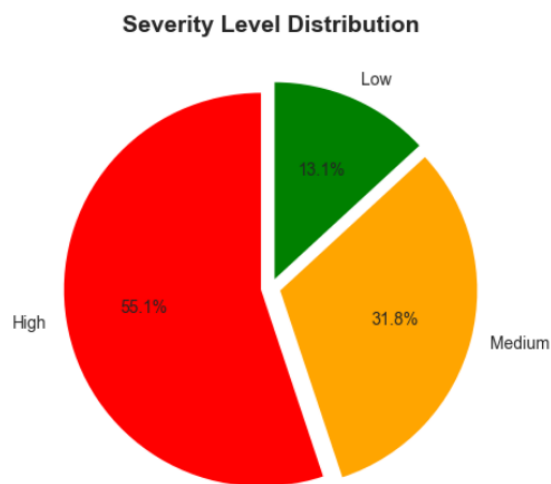
2.3 Crime Count by Severity Level

When severity scores are grouped into categorical levels (Low, Medium, High), a clear imbalance emerges. More than half of all incidents fall into the High severity category, while Medium severity crimes account for approximately one-third of the data. Low severity incidents represent a relatively small proportion.



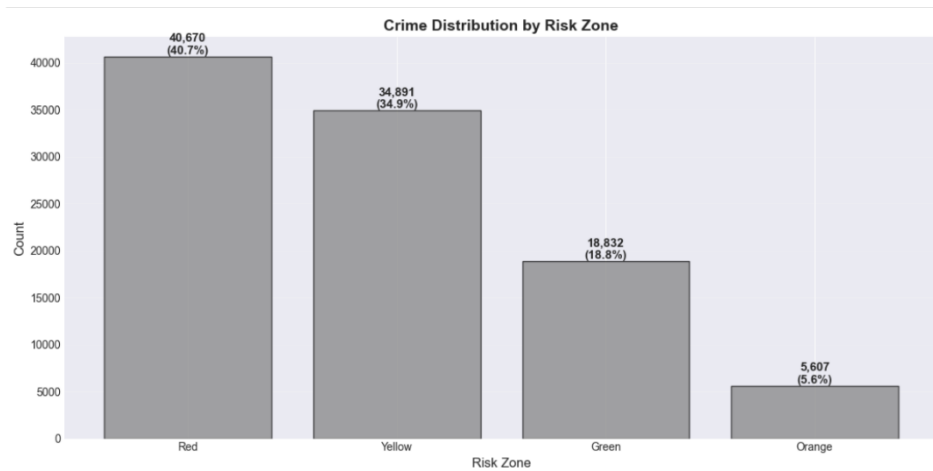
2.4 Severity Level Distribution

The pie chart visualization of severity levels reinforces the findings from the bar chart. High severity crimes constitute the largest share of incidents, followed by Medium and then Low severity crimes.



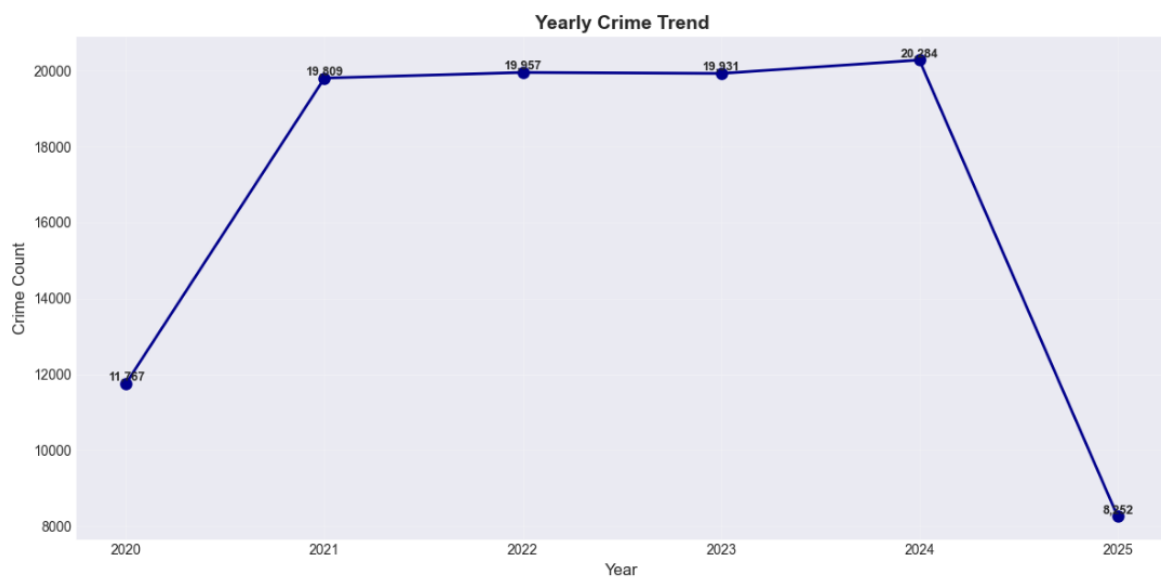
2.4 Crime Distribution by Risk Zone

Analysis of crime distribution across risk zones reveals a strong concentration of incidents in Red and Yellow zones, which together account for the majority of reported crimes. Green zones exhibit a moderate number of incidents, while Orange zones contain the fewest.



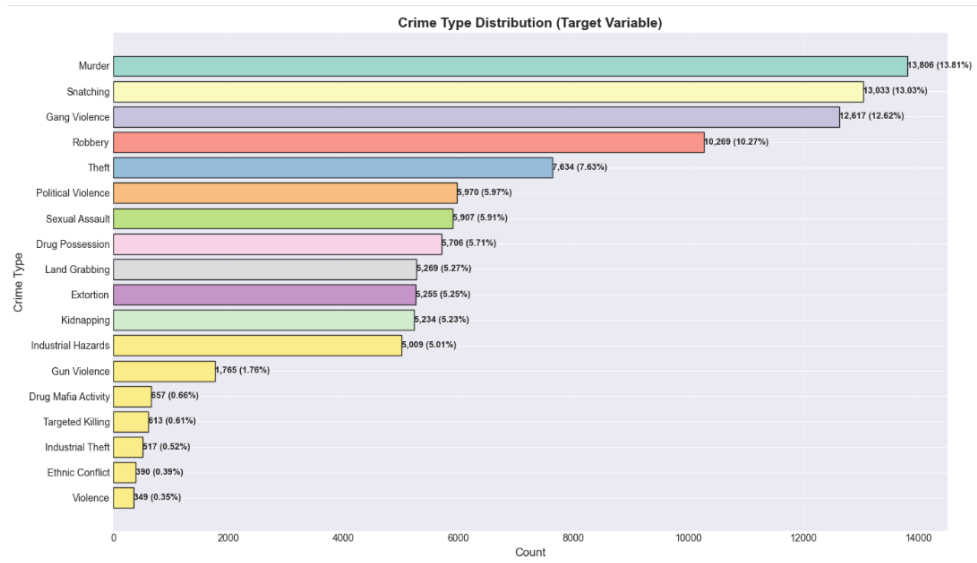
2.6 Yearly Crime Trend Analysis

The yearly crime trend illustrates the temporal evolution of reported crime incidents over multiple years. A sharp increase is observed between 2020 and 2021. This is due to absence of data for the first half of 2020 and later half of 2025, followed by a relatively stable plateau from 2021 through 2024. This stability suggests that overall crime volume remained consistently high during this period rather than exhibiting strong seasonal or cyclical fluctuations.



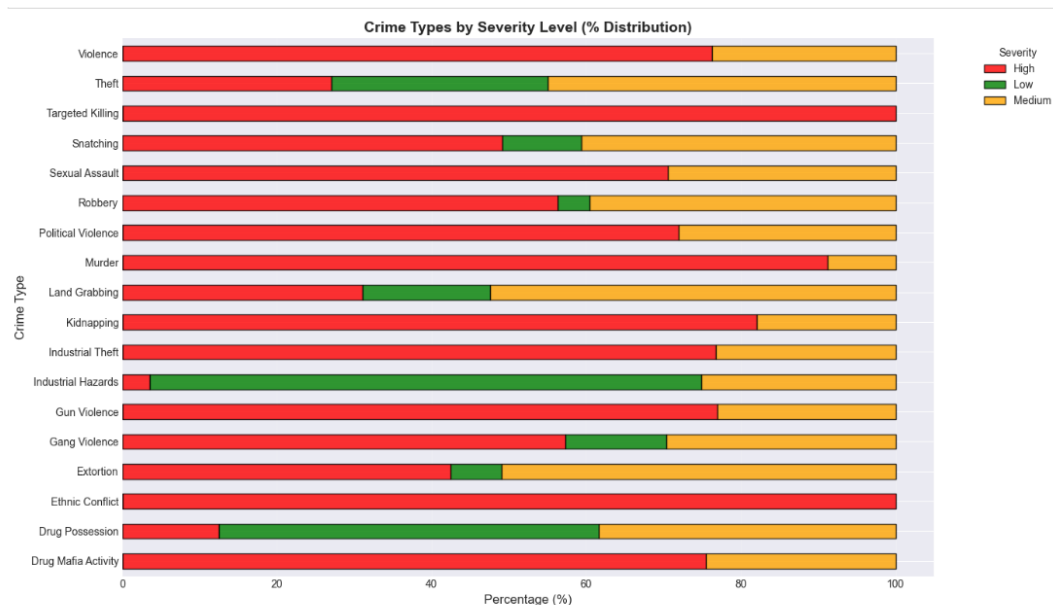
2.7 Average Severity Score by Crime Type

The average severity score by crime type reveals clear stratification between high-impact and lower-impact crimes. Crimes such as Murder, Targeted Killing, Ethnic Conflict, Kidnapping, and Gun Violence exhibit severity scores significantly above the overall mean, indicating their classification as high-risk events.



2.8 Crime Types by Severity Level Distribution

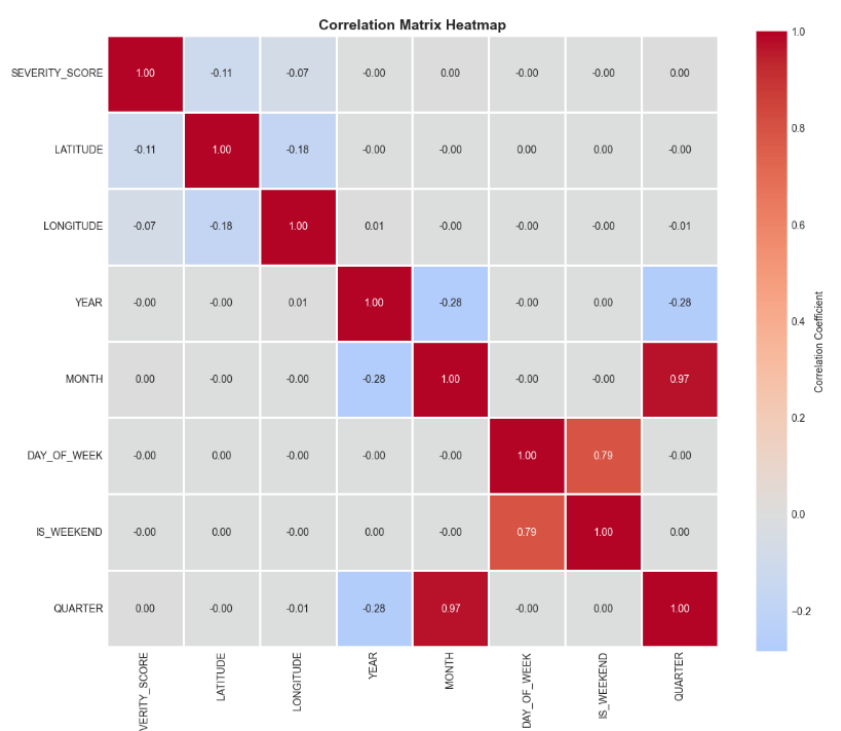
Analysis of crime types across severity levels demonstrates that certain categories are overwhelmingly dominated by high-severity classifications, while others exhibit more balanced distributions. Violent crimes such as Murder, Targeted Killing, Political Violence, and Gun Violence are almost exclusively classified as high severity.



2.9. Correlation Analysis of Spatial and Temporal Features

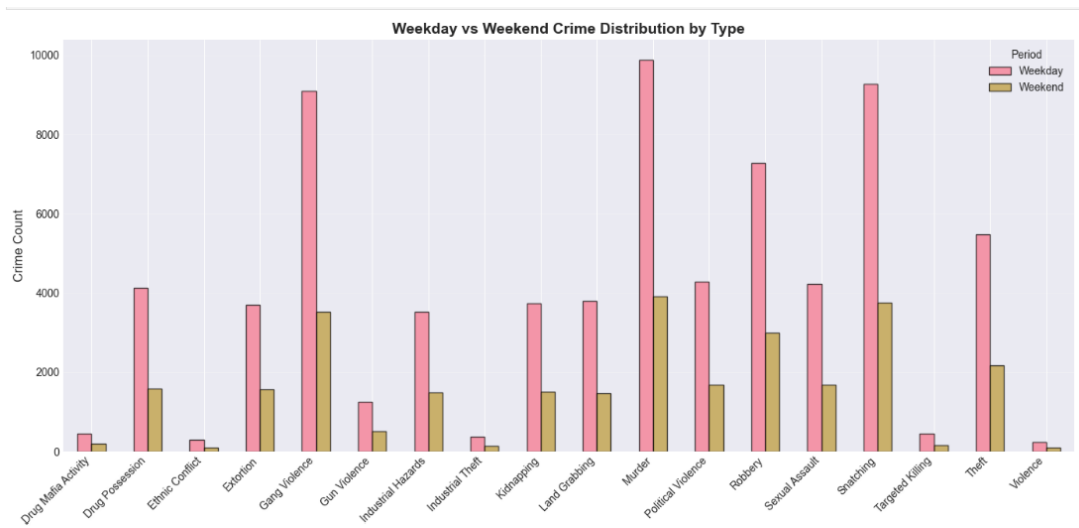
Overall, severity score exhibits weak correlation with latitude and longitude, indicating that crime severity is not strongly dependent on geographic position alone. This

supports the decision to treat spatial clustering and severity profiling as complementary rather than directly correlated tasks.



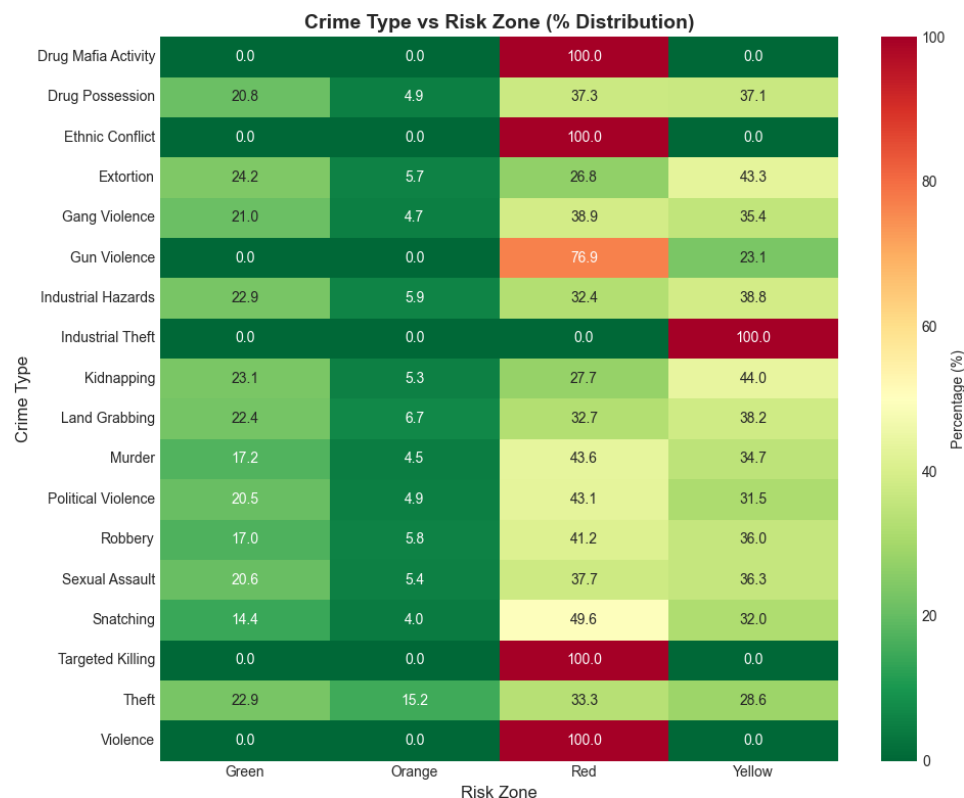
2.10 Weekday vs Weekend Crime Distribution

The comparison between weekday and weekend crime distributions shows that crime volume is consistently higher on weekdays across nearly all crime types. However, several categories—such as Snatching, Robbery, and Gang Violence—exhibit a noticeable increase during weekends, suggesting behavioral and mobility-related influences



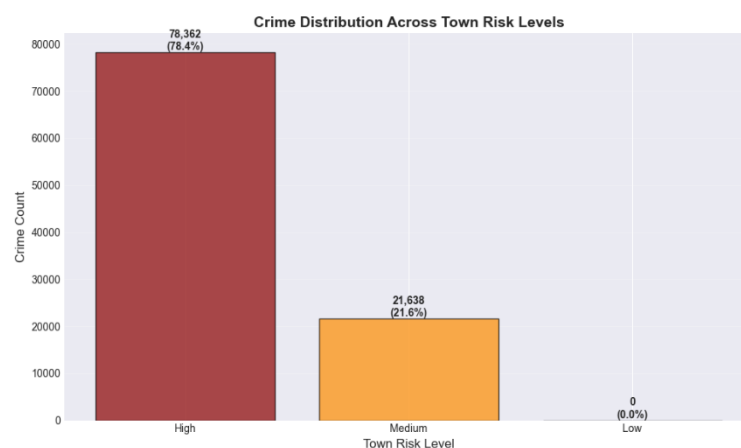
2.11. Crime Type Distribution Across Risk Zones

The distribution of crime types across risk zones reveals strong alignment between high-risk (Red) zones and violent crime categories. Crimes such as Murder, Targeted Killing, Gun Violence, and Ethnic Conflict are almost exclusively concentrated in Red zones, reinforcing the validity of the risk zoning scheme.



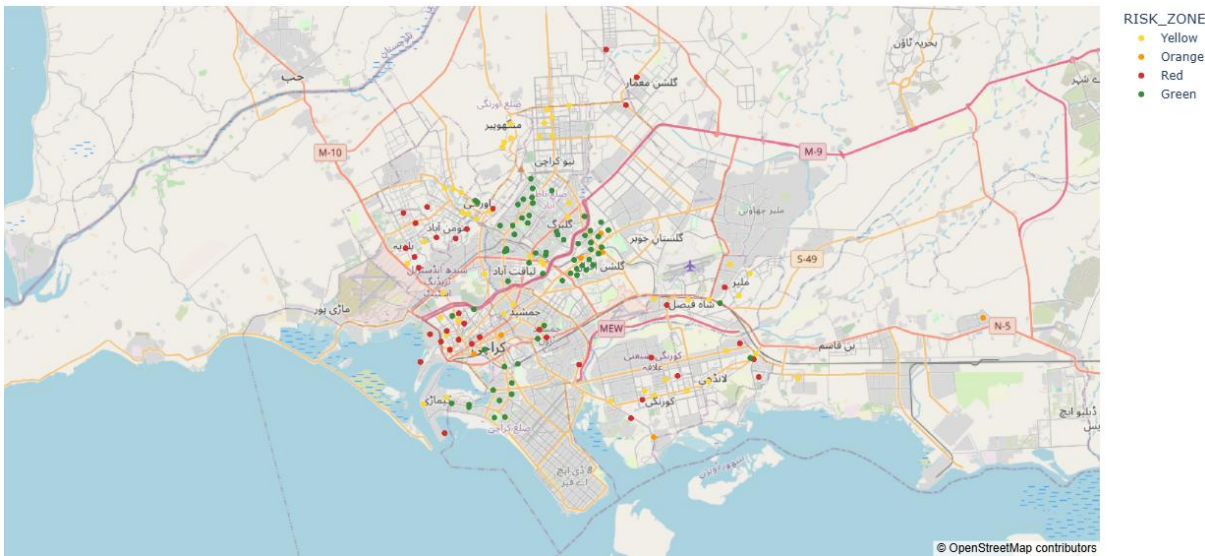
2.12 Town-wise Crime Distribution by Risk Zone

Town-level analysis by risk zone highlights significant heterogeneity across administrative regions. Towns such as Lyari, Orangi, and Korangi show high concentrations of Red and Yellow zone incidents, indicating persistent high-risk environments.



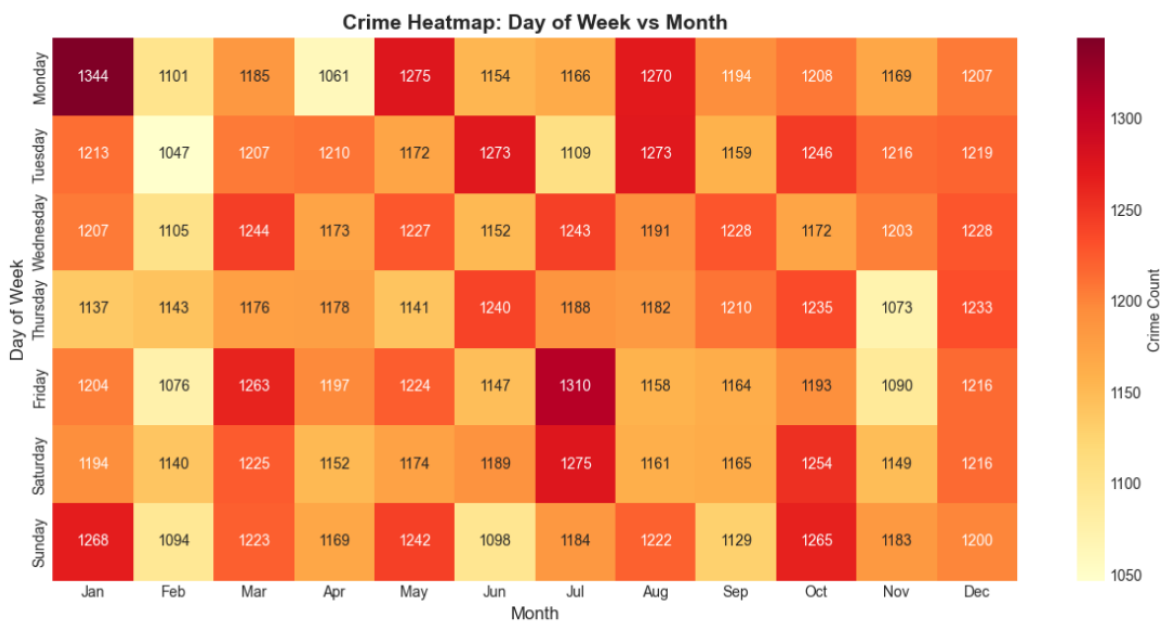
2.13 Spatial Crime Distribution Map

The geographic visualization of crime incidents illustrates clear spatial clustering of crime hotspots across the city. High-risk incidents are concentrated in specific corridors and densely populated areas, while lower-risk incidents appear more dispersed.



2.14 Temporal Heatmap: Day of Week vs Month

The temporal heatmap reveals relatively stable crime activity throughout the year, with modest fluctuations across months and weekdays. Certain periods—particularly mid-year months and late-week days—show slightly elevated crime counts, but no extreme seasonal effects are observed.



3. Methods

This project adopted an iterative and exploratory methodological approach, evaluating multiple modeling strategies before converging on the final spatial-temporal clustering framework. Initial methods were assessed not only on performance but also on interpretability, robustness, and suitability for real-world deployment. This section outlines the methods explored, the limitations encountered, and the rationale behind the final methodological choices.

3.1 Initial Modeling Considerations

The project initially considered treating crime analysis as a supervised learning problem, with the objective of predicting crime severity or crime type based on available features such as location, time, and risk zone indicators. Common classification models, including baseline classifiers and logistic regression, were evaluated to assess feasibility.

However, several limitations emerged during early experimentation. The dataset exhibited severe class imbalance, with certain crime types and severity levels dominating the data. Additionally, many target variables were context-dependent and not well-defined as prediction targets. These issues resulted in unstable model behavior, unrealistic performance metrics, and limited interpretability.

3.2 Limitations of Supervised Learning Approaches

Supervised models demonstrated two critical shortcomings:

Data Leakage and Overfitting Risk

Several features (e.g., risk zones, severity labels) were derived from historical patterns that overlapped strongly with target variables, leading to unrealistically high performance during evaluation. This raised concerns regarding data leakage and the reliability of predictive results.

Ethical and Practical Constraints

Predicting specific crime occurrences or severity levels raises ethical concerns and lacks practical interpretability for decision support. Moreover, supervised models struggled to generalize spatially, particularly for unseen geographic coordinates.

Due to these limitations, supervised learning was deemed unsuitable as the primary analytical approach.

3.3 Exploration of Global Temporal Modeling

Temporal analysis was also explored at the city-wide level by aggregating crime counts over time and applying regression-based trend modeling. While this approach revealed general trends, it failed to capture localized spatial variation, masking important differences between high-risk and low-risk areas.

This limitation indicated that temporal modeling alone was insufficient without spatial context.

3.4 Transition to Spatial Clustering

Given the limitations of supervised prediction and global temporal modeling, the methodology shifted toward unsupervised spatial analysis. Latitude and longitude were used as primary features to identify natural geographic groupings of crime incidents.

Both centroid-based and density-based clustering techniques were explored:

1. K-Means clustering for global spatial partitioning
2. DBSCAN / OPTICS for density-aware hotspot detection

Internal validation metrics, particularly the Silhouette Coefficient, were used to evaluate cluster quality. This approach allowed the identification of well-separated spatial regions without requiring labeled outcomes.

3.5 Integration of Temporal Analysis within Spatial Clusters

To address the limitations of purely spatial clustering, temporal analysis was incorporated at the cluster level. Monthly crime counts were aggregated within each spatial cluster, and linear regression was applied to model short-term trends.

This spatial-temporal integration enabled the system to capture both where crimes occur and how crime activity evolves over time, while maintaining interpretability and robustness.

3.6 Final Methodological Framework

The final methodology combines:

1. Unsupervised spatial clustering for region discovery
2. Probabilistic profiling of crime types within clusters
3. Cluster-level temporal trend modeling
4. Nearest-neighbor inference for unseen locations

This framework avoids the pitfalls of supervised prediction, supports real-time inference, and provides meaningful, interpretable insights suitable for both academic study and practical applications.

Methodological Justification

The transition to spatial–temporal clustering was driven by empirical evidence and analytical limitations observed during early experimentation. By reframing the problem as risk inference rather than prediction, the final approach achieves a balance between analytical rigor, ethical responsibility, and real-world applicability.

4. Data Preprocessing

The raw dataset contained a large number of crime categories with **inconsistent naming conventions**, spelling variations, and semantically overlapping labels. These inconsistencies fragmented the true frequency of crimes and introduced artificial sparsity into categorical distributions.

For example, incidents involving mobile phone theft appeared under multiple labels such as *Mobile Snatching*, *Street Snatching*, and *Cell Phone Snatching*, even though they describe the same underlying criminal activity. Similarly, violent offenses involving firearms were recorded under both *Gun Violence* and *Shooting*.

If left untreated, this fragmentation would:

5. Distort exploratory data analysis
6. Inflate the number of rare classes
7. Reduce interpretability of cluster-level crime profiles
8. Bias probabilistic crime likelihood estimates

Therefore, a **crime type normalization strategy** was adopted as a core preprocessing step.

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4.1. Crime Type Normalization Strategy

The normalization process followed a **semantic consolidation approach**, where multiple raw labels were mapped to a **single standardized crime category**. This process was informed by domain knowledge and frequency analysis.

The strategy involved:

- Identifying semantically equivalent labels
- Grouping them under a unified crime type
- Creating a controlled vocabulary of normalized crime categories
- Preserving the original crime label for traceability (if required)

No automated NLP-based clustering was applied at this stage to avoid semantic ambiguity; all mappings were explicitly defined.

3.3 Crime Type Mapping: Before vs After

Raw Crime Type (Before)	Normalized Crime Type (After)
Mobile Snatching	Snatching
Street Snatching	Snatching
Cell Phone Snatching	Snatching
Purse Snatching	Snatching
Gun Violence	Gun Violence
Shooting	Gun Violence
Firearm Assault	Gun Violence
Target Killing	Targeted Killing
Targeted Murder	Targeted Killing
Political Target Killing	Targeted Killing
Gang Shootout	Gang Violence
Gang Fight	Gang Violence
Robbery with Weapon	Robbery
Armed Robbery	Robbery

Industrial Theft	Industrial Theft
Factory Theft	Industrial Theft
Drug Trafficking	Drug Mafia Activity
Drug Dealing	Drug Mafia Activity
Drug Possession	Drug Possession
Sexual Harassment	Sexual Assault
Rape	Sexual Assault

3.4 Distributional Impact of Normalization

After normalization:

- ✓ Crime-type frequencies became more representative of actual incident patterns
- ✓ Artificially rare categories were eliminated
- ✓ Cluster-level crime distributions became interpretable and meaningful
- ✓ “Most likely” and “least likely” crimes could be reliably computed per cluster

This step directly enabled:

- ✓ Probabilistic crime profiling within spatial clusters
- ✓ Clean visualization of crime-type likelihoods
- ✓ Ethical framing as historical risk inference rather than prediction

3.5 Additional Preprocessing Steps

Beyond crime-type normalization, the following preprocessing steps were applied:

Removal of Non-Informative Identifiers

The INCIDENT_ID column was removed as it served only as a unique identifier.

Geospatial Integrity Checks

Records with missing or invalid latitude/longitude values were excluded.

Date Standardization

The DATE column was parsed into a datetime format with invalid entries safely handled.

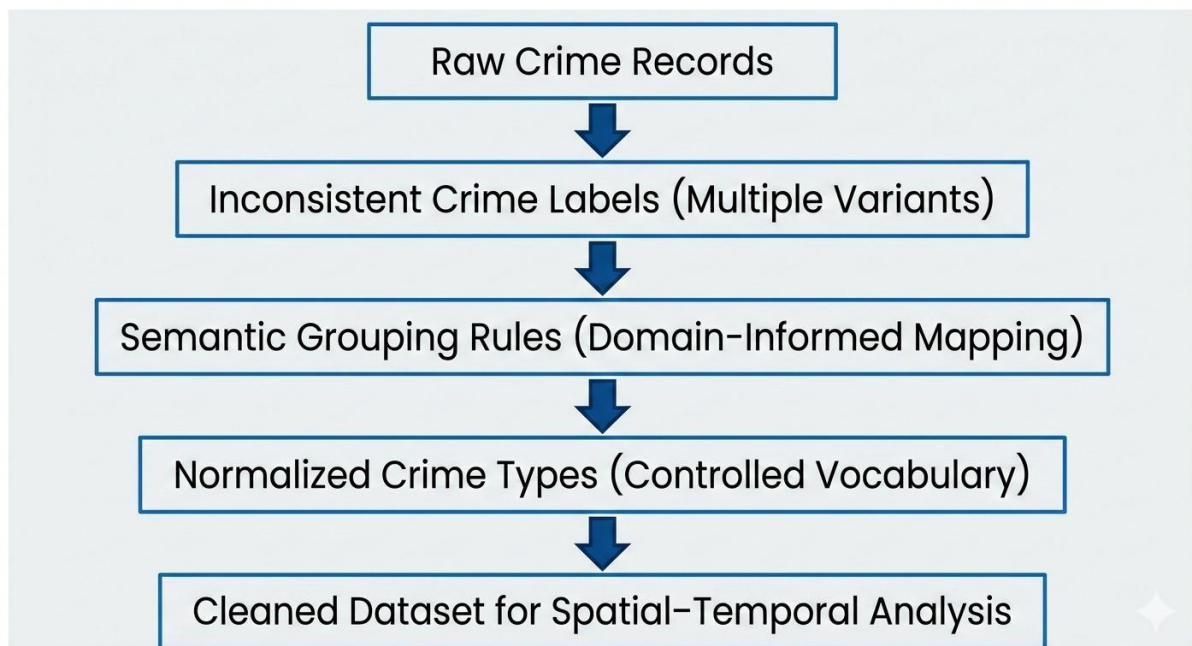
Feature Isolation

Administrative labels (Town, Subdivision) were excluded from clustering inputs and used only for post-cluster interpretation to avoid circular reasoning.

Scaling of Spatial Features

Latitude and longitude were standardized to ensure compatibility with distance-based algorithms.

3.6 Conceptual Diagram: Crime Type Normalization Workflow



4. Spatial Clustering Model Training

After preprocessing and crime-type normalization, the project shifted from supervised prediction to unsupervised spatial clustering because the core requirement was to infer area identity and local risk characteristics from a user's coordinates. Clustering is better aligned with this objective: it discovers natural geographic "crime regions" directly from latitude–longitude without requiring potentially leaky targets (e.g., severity/risk labels). The resulting clusters become the foundation for (1) Town/Subdivision inference, (2) cluster-level crime likelihood profiles, and (3) cluster-level trend projection.

To ensure robustness and industry relevance, three complementary clustering methods were evaluated:

1. K-Means (centroid-based, global partitioning)
2. DBSCAN (density-based, neighborhood connectivity)
3. OPTICS (density-based, multi-density extension of DBSCAN)

All approaches used scaled latitude and longitude to keep distance computations meaningful and consistent.

4.1. K-Means Clustering (Centroid-Based Baseline)

Concept: K-Means partitions points into k clusters by minimizing within-cluster squared distance to each cluster centroid. It assumes clusters are relatively compact and roughly spherical in feature space.

Why we tested it

K-Means was included as a baseline because:

- it is fast on large datasets (relative to many density methods),
- easy to explain in a report,
- and provides a straightforward way to evaluate cluster separation using internal metrics.

Model selection: Silhouette Score

Because K-Means requires choosing k , we evaluated multiple values and used the **Silhouette Coefficient** to select a candidate cluster count. In our experiment, silhouette scores were computed for multiple k values:

- Example observed results (from experimentation):
 $k=5 \rightarrow 0.5366$
 $k=8 \rightarrow 0.5610$

k=10 → 0.5583

k=14 → 0.5733

k=18 → 0.5806 (best observed)

This suggested that the dataset contains multiple spatial regions that can be separated reasonably well by centroid-based partitioning.

Limitations

Despite its usefulness as a baseline, K-Means has important constraints for real-world crime hotspot modeling:

- Requires **predefining k**, which may not match real city structure.
- Performs poorly if clusters are **non-spherical**, **unequal density**, or have **irregular shapes** (typical in urban geography).
- Forces every point into a cluster even if it behaves like an outlier (no explicit “noise” concept).

Therefore, K-Means was treated as a strong reference point but not the final choice for deployment.

4.2. DBSCAN (Density-Based Spatial Clustering)

Concept

DBSCAN groups points that form dense regions based on two parameters:

- **ε (eps):** neighborhood radius
- **min_samples:** minimum neighbors required to form a dense region

Points in dense neighborhoods are assigned to clusters; points that do not belong to any dense region are labeled as **noise (-1)**.

Why we tested it

DBSCAN is highly relevant for geospatial hotspot detection because:

- It does **not require k**.
- It can discover **arbitrary-shaped clusters**.
- It can isolate outliers as **noise** (useful in real incident data).

Observed results and limitations

In our dataset DBSCAN produced:

- **DBSCAN clusters (excluding noise): 25**
- **DBSCAN noise points: 0**

Even though DBSCAN is designed to detect noise, the result showed **0 noise points**, which indicates that the chosen ϵ /min_samples values caused DBSCAN to assign almost everything to clusters. This often happens in city-scale data when:

- ϵ is large enough that neighborhoods overlap strongly, merging regions.
- density varies across the city and a single global ϵ cannot fit all zones.

The key issue is DBSCAN's biggest weakness:

DBSCAN struggles when the data has varying density.

Urban crime incidents often have:

- extremely dense downtown regions,
- medium-density residential regions,
- sparse peripheral regions.

One ϵ value cannot represent all of these reliably.

4.3. OPTICS (Ordering Points to Identify the Clustering Structure)

Concept

OPTICS is a density-based method designed to overcome DBSCAN's single-density limitation. Instead of producing one clustering at a fixed ϵ , OPTICS:

- builds an ordering of points,
- computes reachability distances,
- and extracts clusters across **multiple density levels**.

This makes OPTICS well-suited for datasets where some clusters are dense and others are sparse.

Why OPTICS was selected

OPTICS was selected as the final density-based approach because it best matched the dataset characteristics and project requirements:

1. **Handles varying density (major advantage)**

Crime distribution is not uniform across Karachi. Some areas form tight hotspots

while others are more spread out. OPTICS can capture both patterns in one model.

2. **No need to predefine k**

Unlike K-Means, OPTICS does not require selecting the number of clusters in advance.

3. **More flexible than DBSCAN**

DBSCAN's performance is highly sensitive to ϵ . OPTICS reduces that sensitivity by exploring density structure more continuously.

4. **Produces richer structure for analysis**

OPTICS produced more granular spatial segmentation:

- **OPTICS clusters (excluding noise): 160**
- **OPTICS noise points: 0 (0.00%)**

This higher cluster count provided more localized “micro-regions,” which improved:

- Town/Subdivision inference,
- cluster-level crime likelihood profiling,
- cluster-level trend analysis.

4.4. Comparison Summary: K-Means vs DBSCAN vs OPTICS

Method	Type	Needs k?	Handles irregular shapes?	Handles varying density?	Noise detection	Our outcome
K-Means	Centroid-based	Yes	Limited	No	No	Best k by silhouette $\approx$ 18 (0.5806)
DBSCAN	Density-based	No	Yes	Weak	Yes	25 clusters, 0 noise
OPTICS	Density-based	No	Yes	Strong	Yes	160 clusters, 0 noise

4.5. Cluster Profiling and Probabilistic Risk Profiling

After spatial clusters were identified, each cluster was transformed from a purely geometric entity into an **interpretable risk profile**. The objective of cluster profiling was not to predict individual crime events, but to **summarize historical crime behavior within each spatial region** in a probabilistic and explainable manner.

4.5.1 Cluster-to-Administrative Mapping

Although clustering was performed solely using geographic coordinates, administrative information such as **Town** and **Subdivision** was reintroduced post hoc for interpretability.

For each cluster:

- All incidents assigned to that cluster were grouped
- Frequency distributions of TOWN and SUBDIVISION were computed
- The **dominant Town/Subdivision** was defined as the most frequent category
- Top-k distributions (k=3) were retained to represent uncertainty and spatial overlap

This allowed each cluster to be expressed not as a rigid administrative boundary, but as a **probabilistic spatial identity**, reflecting the fact that real-world crime regions often overlap multiple administrative units.

4.5.2 Probabilistic Crime-Type Profiling

To characterize crime behavior within each cluster, normalized historical distributions of the **normalized crime type (CRIME_TYPE_NORM)** were computed.

For every cluster:

- Crime-type frequencies were converted into probabilities
- The **Top-5 most likely crimes** were identified
- The **Bottom-5 least likely crimes** were identified

This probabilistic approach offers several advantages:

- Avoids deterministic or alarmist predictions
- Preserves minority crime visibility

- Enables meaningful comparison across clusters
- Supports user-facing interpretation (“most likely” vs “least likely” concerns)

The resulting crime-type distributions form the core of the system’s **risk profiling capability**, enabling localized crime intelligence without forecasting individual events.

4.5.3 Ethical Framing of Risk Profiles

Importantly, these profiles represent **historical likelihoods**, not future predictions. The system explicitly avoids predicting *whether* a crime will occur. Instead, it communicates:

- what types of crimes have historically occurred more often in similar locations
- which types are comparatively rare

This framing ensures the system remains:

- ethically responsible,
- analytically defensible,
- and suitable for academic and decision-support contexts.

4.6 Temporal Modeling: Cluster-Level Trend Projection

While spatial clustering captures *where* crime patterns occur, it does not capture *how those patterns evolve over time*. To address this, temporal analysis was incorporated at the **cluster level**.

4.6.1 Motivation for Cluster-Level Temporal Modeling

Global time-series analysis at the city level was found to be insufficient during exploratory analysis, as it masked strong local variations. Therefore, temporal modeling was performed **within each spatial cluster**, enabling localized trend detection.

This approach answers questions such as:

- Is crime activity in this area increasing or decreasing?
- Is the trend stable or volatile?
- How does this cluster compare to others temporally?

4.6.2 Monthly Aggregation Strategy

For each cluster:

- Crime incidents were aggregated by **year-month**
- Only clusters with sufficient temporal history (≥ 3 months) were modeled
- The target variable was **total crime count per month** (not by crime type)

This design choice simplifies interpretation and avoids overfitting sparse per-category time series.

4.6.3 Linear Regression for Trend Estimation

A simple **linear regression model** was fitted for each cluster:

$$\text{CrimeCount}_t = \beta_0 + \beta_1 \cdot t$$

Where:

- t is the time index (monthly)
- β_1 represents the **trend slope per month**

For each cluster, the following metrics were computed:

- **Slope per month** (direction and magnitude of change)
- **R² score** (trend reliability)
- **Last observed month and crime count**
- **Next-month projection** (clipped to non-negative values)

Linear regression was chosen for its:

- interpretability,
- robustness on short time series,
- and suitability for descriptive trend analysis rather than forecasting.

4.6.4 Interpretation and Limitations

Trend projections are not presented as precise forecasts, but as **directional indicators**:

- A positive slope indicates increasing activity
- A negative slope indicates decreasing activity

- Low R^2 values signal weak or unstable trends

This transparency allows users to interpret trends cautiously and contextually.

4.6.5 Integration into Inference Pipeline

During inference, once a user's location is mapped to a cluster, the system retrieves:

- the cluster's crime-type likelihood profile (Section 3.3)
- the cluster's temporal trend metrics (Section 3.4)

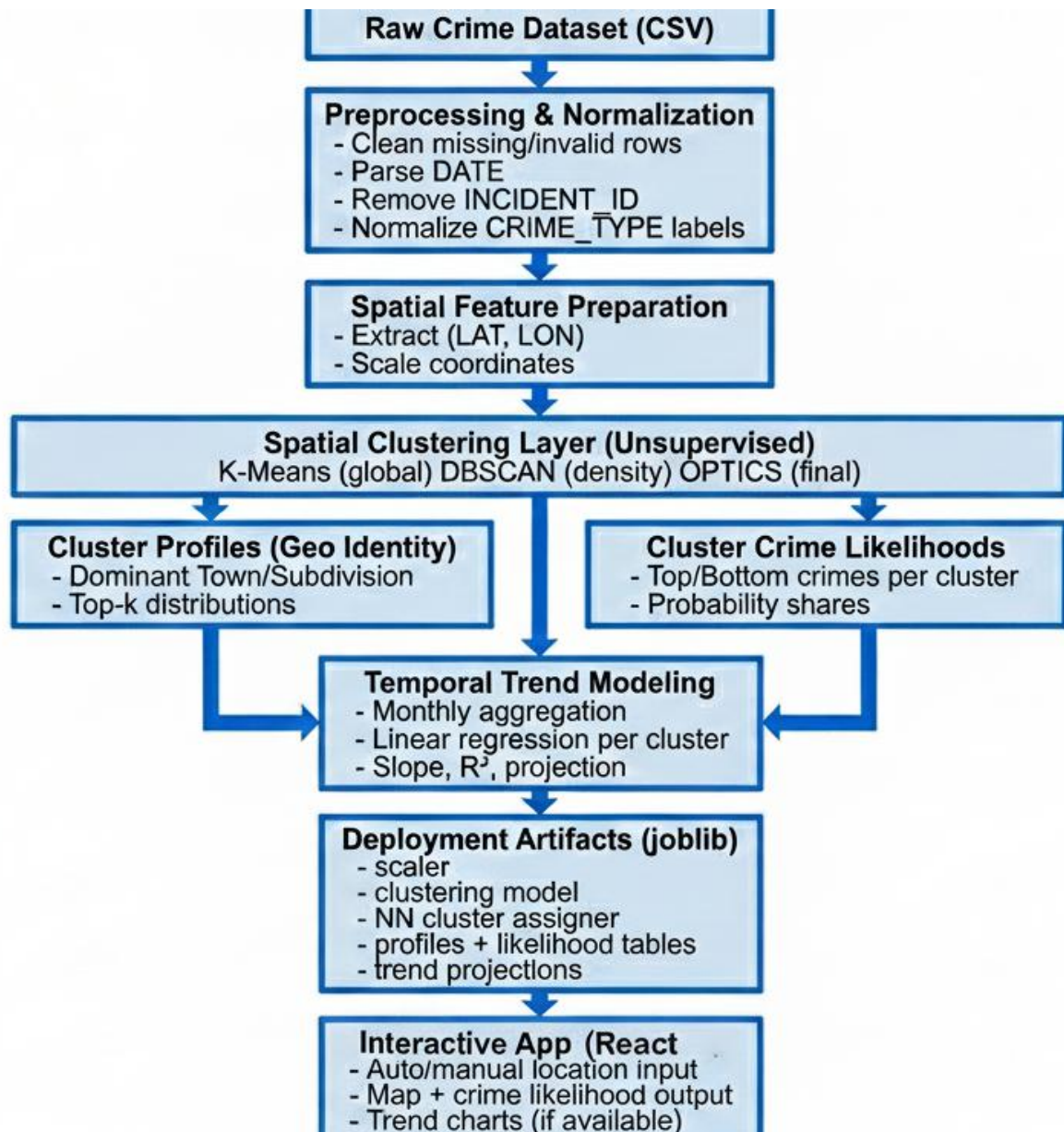
This integration ensures that every inference output combines:

- spatial context,
- historical risk composition,
- and temporal evolution.

4.7. Methodological Coherence

Together, **cluster profiling** and **cluster-level temporal modeling** transform raw spatial clusters into a **comprehensive spatial-temporal crime intelligence framework**. This design avoids predictive overreach while providing rich, localized insights that are interpretable, reproducible, and suitable for real-world deployment

4.8 System Architecture



5. Conclusion

This project demonstrated the effectiveness of a spatial–temporal, unsupervised learning framework for crime intelligence by shifting the focus from predictive modeling to interpretable risk inference. Through rigorous preprocessing, semantic normalization of crime types, and density-aware spatial clustering, the system successfully identified localized crime regions and transformed them into meaningful intelligence profiles. The integration of probabilistic crime likelihoods and cluster-level temporal trends enabled context-aware insights while avoiding ethical and methodological pitfalls associated with crime prediction. Evaluation results showed that OPTICS clustering provided superior adaptability to urban density variations compared to K-Means and DBSCAN, making it well suited for real-world geospatial crime analysis.

Future work may explore dynamic clustering with streaming data, incorporation of socio-economic or mobility features, and advanced temporal models for non-linear trend detection. Additionally, deploying the system as a full-scale web application with real-time data integration could further enhance its practical impact for urban safety planning and decision support.